Homework 1 (Due: March 6th)

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1) In this exercise, we show how to equip the set $\operatorname{Spec} A = \{\text{all prime ideals of } A\}$ with topology, where A is any commutative ring with unit 1.

i.e. subring of r that is an tve subgroup and $r \in R$, $i \in I \Rightarrow ri, ir \in I$

Let I, J be ideals of A. Define $V(S) := \{p \in \operatorname{Spec} A \mid S \subseteq p\}$ for any subset $S \subseteq A$. For $f \in A$, define $D(f) := \{p \in \operatorname{Spec} A \mid f \notin p\}$.

Prove the following:

(i) $V(I) \cup V(J) = V(I \cap J)$

$\forall P \in V(I \cap J)$, by def, $P \in \operatorname{Spec} A$ and $I \cap J \subseteq P \Leftrightarrow P \in \operatorname{Spec} A$ and $(I \subseteq P$ or $J \subseteq P)$

$\Leftrightarrow (P \in \operatorname{Spec} A$ and $I \subseteq P)$ or $(P \in \operatorname{Spec} A$ and $J \subseteq P)$

$\Leftrightarrow$ By def, $P \in V(I)$ or $P \in V(J)$

$\Leftrightarrow P \in V(I) \cup V(J)$

$\therefore V(I) \cup V(J) = V(I \cap J)$. $\square$

(ii) Let $(I_\lambda)_{\lambda \in \Lambda}$ be a family of ideals of A. Then, $\bigcap_\lambda V(I_\lambda) = V(\sum_\lambda I_\lambda)$.

"$\subseteq$": $\forall P \in \bigcap_\lambda V(I_\lambda)$, by def, $\forall \lambda$, $I_\lambda \subseteq P$ and $P \in \operatorname{Spec} A$.

Also, $\forall x \in \sum_\lambda I_\lambda$, $\exists x_1, \dots, x_k$, s.t. $x_i \in I_{\lambda_i}$ $\forall i$, $x_1 + \dots + x_k = x$.

$\therefore \forall i$, $x_i \in I_{\lambda_i} \subseteq P \Rightarrow x_1 + \dots + x_k = x \in P$

$\therefore \sum_\lambda I_\lambda \subseteq P$ and $P \in \operatorname{Spec} A$, so by def, $P \in V(\sum_\lambda I_\lambda)$ ✓

"$\supseteq$": $\forall P \in V(\sum_\lambda I_\lambda)$, by def, $P \in \operatorname{Spec} A$ and $\sum_\lambda I_\lambda \subseteq P$

$\because \forall \lambda$, $I_\lambda \subseteq \sum_\lambda I_\lambda$, by def

$\therefore I_\lambda \subseteq P$ (and $P \in \operatorname{Spec} A$) $\forall \lambda$

$\therefore$ By def, $P \in V(I_\lambda)$ $\forall \lambda \Rightarrow P \in \bigcap_\lambda V(I_\lambda)$ ✓

$\therefore \bigcap_\lambda V(I_\lambda) = V(\sum_\lambda I_\lambda)$ $\square$

(iii) $V(A) = \emptyset$ and $V(0) = \operatorname{Spec} A$

- "$V(A) = \emptyset$": Assume not, i.e. $\exists P \in V(A)$.

  $\therefore$ By def, $P \in \operatorname{Spec} A$ and $A \subseteq P$. As $A \subseteq P \Rightarrow P = A$, but $A \notin \operatorname{Spec} A$, this is a contradiction ⚡

  $\therefore$ By contradiction, $V(A) = \emptyset$. $\square$

- "$V(0) = \operatorname{Spec} A$": By def, $V(0) = \{P \in \operatorname{Spec} A \mid \{0\} \subseteq P\}$.

  However, as $0 \in I$ $\forall I$: ideal of A, so $0 \in P$ $\forall P \in \operatorname{Spec} A$

  $\therefore V(0) = \{P \in \operatorname{Spec} A \mid P\} = \operatorname{Spec} A$. $\square$

Remark

(1) This exercise shows that there is a unique topology on $\operatorname{Spec} A$ whose closed subsets are of the form $V(I)$ for an ideal I of A, and $D(f)$ form a basis of open subsets on $\operatorname{Spec} A$. This topology is called the Zariski topology.

(2) For a maximal ideal m in A, $V(m) = \{m\}$, hence a maximal ideal corresponds to a closed point of $\operatorname{Spec} A$.

2) Let $A = k[x]$, where k is a field. Find all closed points of $\operatorname{Spec} A$.

In other words, we need to find all maximal ideals of $A = k[x]$

As $k[x]$ is ED, it must be a PID,

$\therefore$ By def, all ideals of $A = k[x]$ are of the form (a), $a \in A$.

As A: domain, thus (0) is prime.

Also, a nonzero ideal (f) is prime $\Leftrightarrow f$: irred

$\therefore \operatorname{Spec} A = \{(f) \mid f = 0$ or f: irred, $f \in k[x]\}$

As $V(0) = \operatorname{Spec} A \neq 0$, thus (0) is not a closed point of $\operatorname{Spec} A$.

Fix any f: irred, consider $V((f)) = \{P \in \operatorname{Spec}(k[x]) \mid (f) \subseteq P\}$

(From above) $= \{(g) \mid (f) \subseteq (g)$ and g: irred, $g \in A\}$

If $\exists (g) \in V((f))$, $g \neq f$, then $(f) \subseteq (g) \Rightarrow f = hg$ for some $h \in A$, $h \neq 1$, but f is irred ⚡

$\therefore$ By contradiction, thus $V((f)) = (f)$

$\therefore$ The closed points of $\operatorname{Spec} A$ are $\{(f) \mid f \in A, f$: irred$\}$.

3) Let $A=\mathbb{Q}[x,y]/(x^2-y^3)$. Show that A is an integral domain but not integrally closed. Find the integral closure of A in $\mathrm{Frac}(A)$. Shun/翔海(@shun4midx)

- <u>A is an integral domain</u>

  By def, we need to prove "$\forall a,b\in A,\ ab=0\Rightarrow a=0$ or $b=0$"

  Claim: x^2-y^3 is irred in $\mathbb{Q}[x,y]$

  <u>Proof</u>: Let $f(x,y)=x^2-y^3$. We can view f as $f\in\mathbb{Q}(y)[x]$

  We see, over $\mathbb{Q}(y)$, f is reducible $\Leftrightarrow \exists$ root in $\mathbb{Q}(y)\Leftrightarrow y^3$ is a square in $\mathbb{Q}(y)$.

  However, if y^3 is a square in $\mathbb{Q}(y)$, then $\exists a,b\in\mathbb{Q}[y]$, s.t. $y^3=(\frac{a}{b})^2\Rightarrow y^3b^2=a^2$, but $\mathbb{Q}[y]$ is a UFD $\rightarrow\!\leftarrow$

  $\therefore f$ is irred over $\mathbb{Q}(y)\Rightarrow$ By Gauss' Lemma, f is irred over $\mathbb{Q}[x,y]$ ($\because \mathbb{Q}[y]$: UFD, f: primitive) ✓

  $\therefore \forall a,b\in\mathbb{Q}[x,y],\ ab=x^2-y^3\Rightarrow a=x^2-y^3$ or $b=x^2-y^3$.

  Moreover, $\forall a,b\in\mathbb{Q}[x,y]$, if $\exists c\in\mathbb{Q}$, s.t. $ab=c(x^2-y^3)$, then $a=c_1(x^2-y^3)$ or $b=c_2(x^2-y^3)$, $c_1,c_2\in\mathbb{Q}$

  In other words, $\forall a,b\in\mathbb{Q}[x,y]/(x^2-y^3)$, $ab=\bar{0}\Rightarrow a=\bar{0}$ or $b=\bar{0}$

  $\therefore$ By def, A is an integral domain. □

- <u>A is not integrally closed</u>

  By def, we need to show $\{f\in\mathrm{Frac}(A)\mid f\text{: integral over } A\}\not\subseteq A$

  In other words, we need to show $\exists f\notin A$, s.t. $f(x,y)=\frac{g(x,y)}{h(x,y)}$, $g,h\in A$, and $\exists$ monic $q(z)\in A[z]$, s.t. $q(f)=0$

  Consider $f:=\frac{x}{y}\in\mathrm{Frac}(A)\setminus A$, $q(z):=z^2-y\in A[z]$, which is monic

  Then, $q(f)=(\frac{x}{y})^2-y=\frac{x^2-y^3}{y^2}$

  As x^2-y^3 is irred in $\mathbb{Q}[x,y]$, thus in $A=\mathbb{Q}[x,y]/(x^2-y^3)$, $q(f)=\frac{0}{y}=0$

  $\therefore$ A is not integrally closed. □

- <u>Integral closure of A in Frac(A)</u>

  By def, $\bar{A}=\{f\in\mathrm{Frac}(A)\mid f\text{: integral over } A\}$, $A=\mathbb{Q}[x,y]/(x^2-y^3)$

  Notice, if $x=t^3$, $y=t^2$, then $x^2-y^3=0$.

  $\therefore A=\mathbb{Q}[x,y]/(x^2-y^3)\cong\mathbb{Q}[t^2,t^3]$, which has integral closure $\mathbb{Q}[t]$ in $\mathbb{Q}(t)$.

  $\therefore \bar{A}=\mathbb{Q}[\frac{x}{y}]$, since $\frac{x}{y}=t$.

4) Let A be an integral domain. Suppose that A is integrally closed. Show that the polynomial ring $A[T]$ is also integrally closed.

By def, $\{x\in\mathrm{Frac}(A)\mid x\text{: integral}/A\}=:\bar{A}\subseteq A$.

Consider $\{f\in\mathrm{Frac}(A[T])\mid f\text{: integral}/A[T]\}=:\widetilde{A[T]}$.

$\forall f\in\widetilde{A[T]}$, we know $\exists g,h\in A[T]$, $h\neq 0$, s.t. $f=\frac{g}{h}$.

If $f\notin A[T]$, then $\frac{g}{h}\notin A[T]$. As $\bar{A}\subseteq A$, thus $\deg h\geq 1$, otherwise $\frac{g}{h}\in A[T]$.

We know $\exists$ monic poly $F(y)\in A[T][y]$, s.t. $F(\frac{g}{h})=0$.

By remainder thm, $\exists Q(y)\in\mathrm{Frac}(A[T])[y]$, $r\in\mathrm{Frac}(A[T])$, s.t. $F(y)=Q(y)\cdot(y-\frac{g}{h})+r$

$F(\frac{g}{h})=Q(\frac{g}{h})\cdot(\frac{g}{h}-\frac{g}{h})+r\Rightarrow r=0$ $\therefore F(y)=(y-\frac{g}{h})Q(y)$

As F: monic, $y-\frac{g}{h}$: monic, thus Q: monic.

$\therefore$ By Gauss' lemma, $y-\frac{g}{h}\in A[T][y]$, so $\frac{g}{h}=f\in A[T]$ $\rightarrow\!\leftarrow$

$\therefore$ By contradiction, $f\in\widetilde{A[T]}\Rightarrow f\in A[T]$, i.e. $\widetilde{A[T]}\subseteq A[T]$

$\therefore$ By def, $A[T]$ is integrally closed. □

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5) Show that if $d \neq 1$ is a square-free integer, then

$$\mathcal{O}_{\mathbb{Q}[\sqrt{d}]} = \begin{cases} \mathbb{Z}[\frac{1+\sqrt{d}}{2}] & \text{if } d \equiv 1 \pmod 4 \\ \mathbb{Z}[\sqrt{d}] & \text{if } d \equiv 2 \text{ or } 3 \pmod 4 \end{cases}$$

Recall: $\mathcal{O}_K = \{\alpha \in K \mid \alpha : \text{integral} / \mathbb{Z}\}$

Consider $\alpha = p + q\sqrt{d} \in \mathcal{O}_{\mathbb{Q}[\sqrt{d}]}$, $p, q \neq 0 \in \mathbb{Q}$

Notice, $\alpha - p = q\sqrt{d} \in \mathcal{O}_{\mathbb{Q}[\sqrt{d}]} \Rightarrow (\alpha - p)^2 = \alpha^2 - 2\alpha p + p^2 = q^2 d \Rightarrow \alpha^2 - 2\alpha p + (p^2 - q^2 d) = 0$

Set $g(x) := x^2 - 2xp + (p^2 - q^2 d)$, we know $g(\alpha) = 0$.

As α: integral $/\mathbb{Z}$, by def, $\exists f(x) \in \mathbb{Z}[x]$, s.t. $f(\alpha) = 0$, where f: monic

$\therefore$ By remainder thm, $\exists q(x) \in \mathbb{Q}[x]$, $a, b \in \mathbb{Q}$, s.t. $f(x) = q(x)g(x) + (ax+b)$

$f(\alpha) = 0$, $g(\alpha) = 0 \Rightarrow 0 = q(\alpha) \cdot 0 + (a\alpha + b) \Rightarrow a = 0, b = 0$, i.e. $f(x) = q(x)g(x)$

As f, g: monic, thus q: monic.

By Gauss' Lemma, $g(x) \in \mathbb{Z}[x]$, so $2p, p^2 - q^2 d \in \mathbb{Z}$.

<u>Case 1: $2p$ = odd</u>

Say $2p = 2m+1$, $m \in \mathbb{Z}$. Then, $p^2 - q^2 d \in \mathbb{Z} \Rightarrow 4(p^2 - q^2 d) \equiv 0 \pmod 4 \Rightarrow 4p^2 \equiv 4q^2 d \equiv 1 \pmod 4$

$\therefore p, q \notin \mathbb{Z}$, $q = \frac{2n+1}{2}$ for some $n \in \mathbb{Z}$.

$\Rightarrow 4q^2 d = (2n+1)^2 d \equiv d \equiv 1 \pmod 4$, $\frac{2m+1}{2} + \frac{2n+1}{2}\sqrt{d} \in \mathcal{O}_{\mathbb{Q}[\sqrt{d}]}$

$\therefore$ When $d \equiv 1 \pmod 4$, $\mathcal{O}_{\mathbb{Q}[\sqrt{d}]} = \mathbb{Z}[\frac{1+\sqrt{d}}{2}]$

<u>Case 2: $2p$ = even</u>

Then, $p \in \mathbb{Z} \Rightarrow q^2 d \in \mathbb{Z}$

Also, d: square-free $\Rightarrow q \in \mathbb{Z}$.

$\therefore$ When $d \equiv 2, 3 \pmod 4$, $\mathcal{O}_{\mathbb{Q}[\sqrt{d}]} = \mathbb{Z}[\sqrt{d}]$. $\square$

6) Let $A \subseteq B$ be an extension of integral domains, $S \subseteq A \setminus \{0\}$ be a multiplicative set. Suppose A' is the integral closure of A in B. Show that $S^{-1}A'$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.

By def, $A' = \{x \in B \mid x : \text{integral} / A\}$

Recall: $S^{-1}A := \{\frac{a}{s} \mid \frac{a_1}{s_1} = \frac{a_2}{s_2} \Leftrightarrow \exists u \in S, \text{ s.t. } (a_1 s_2 - a_2 s_1)u = 0, \text{ for any } a_1, a_2 \in A, s_1, s_2 \neq 0 \in S\}$

Claim: $S^{-1}A' = \{y \in S^{-1}B \mid y : \text{integral} / S^{-1}A\}$

<u>Proof</u>

"$\subseteq$": $\forall \frac{\tilde{a}}{s} \in S^{-1}A'$, we know $\tilde{a} \in B$, $\tilde{a}$: integral $/A$, i.e. $\exists f(x) \in A[x]$, f: monic, s.t. $f(\tilde{a}) = 0$

Notice, write $f(x) := \sum_{i=0}^{n} \alpha_i x^i$, then we know $\sum_{i=0}^{n} \alpha_i (\tilde{a})^i = 0$, $\alpha_i \in A$ $\forall i$

Let $\beta_i := \frac{\alpha_i}{s^{n-i}}$, $g(x) := \sum_{i=0}^{n} \beta_i x^i \in (S^{-1}A)[x]$

Then, $g(\frac{\tilde{a}}{s}) = \sum_{i=0}^{n} \beta_i (\frac{\tilde{a}}{s})^i = \sum_{i=0}^{n} \frac{\alpha_i}{s^{n-i}} \left(\frac{(\tilde{a})^i}{s^i}\right) = \sum_{i=0}^{n} \frac{\alpha_i (\tilde{a})^i}{s^n} = \frac{1}{s^n} \cancelto{0}{\sum_{i=0}^{n} \alpha_i (\tilde{a})^i} = 0$

As $\alpha_n = 1$, thus $\beta_n = 1$, so g: monic

$\because \tilde{a} \in A' \Rightarrow \tilde{a} \in B$

$\therefore \frac{\tilde{a}}{s} \in S^{-1}B$

$\therefore \frac{\tilde{a}}{s} \in \{y \in S^{-1}B \mid y : \text{integral} / S^{-1}A\}$

"$\supseteq$": $\forall \frac{b}{s} \in \{y \in S^{-1}B \mid y : \text{integral} / S^{-1}A\}$, we know $b \in B$, $s \in S$, and $\exists f(x) \in (S^{-1}A)[x]$, f: monic, s.t. $f(\frac{b}{s}) = 0$.

Say $f(x) = \sum_{i=0}^{n} \frac{a_i}{s_i} x^i$, where $\frac{a_n}{s_n} = 1$.

We can rewrite $(\frac{a_i}{s_i})$: and $\frac{b}{s}$ with a common denominator $(\frac{b_i}{t})$: and $\frac{b'}{t}$, where $b_n = t$. Notice, $t, b_i \in A$ $\forall i$.

Then, $f(\frac{b}{s}) = f(\frac{b'}{t}) = 0 \Rightarrow \sum_{i=0}^{n} \frac{b_i}{t} (\frac{b'}{t})^i = 0 \Rightarrow$ Multiplying both sides by t^n, $(b')^n + b_{n-1}(b')^{n-1} + (t b_{n-2})(b')^{n-2} + \ldots + t^{n-1} b_0 = 0$

$\therefore \exists$ monic $g(x) \in A[x]$, s.t. $g(b') = 0 \Rightarrow b'$: integral $/A$. As $b' \in B$, thus $b' \in A'$.

$\therefore \frac{b}{s} = \frac{b'}{t} \in S^{-1}A$

$\therefore$ $S^{-1}A'$ is the integral closure of $S^{-1}A$ in $S^{-1}B$. $\square$

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7) Let $A \subseteq B$ be an integral extension of rings. Let $q \in \operatorname{Spec} B$, $p = q \cap A \in \operatorname{Spec} A$. Show that q is a maximal ideal of B if and only if p is a maximal ideal of A.

Recall: "I is max $\Leftrightarrow$ B/I is a field" and "I is prime $\Leftrightarrow$ B/I is an integral domain"

Claim: B is a domain, then B: field $\Leftrightarrow$ A: field

<u>Proof</u>

"$\Rightarrow$": $\forall 0 \neq x \in A \subseteq B$, thus $\exists\, x^{-1} = b \in B$, so by def, $\exists$ monic irred $f(T) \in A[T]$, s.t. $f(b) = 0$.

Say $f(T) = \sum_{i=0}^{n} \alpha_i T^i$ where $\alpha_n = 1$, $\alpha_i \in A\ \forall i$.

$f(b) = 0 \Rightarrow b^n + \alpha_{n-1} b^{n-1} + \ldots + \alpha_1 b + \alpha_0 = 0$

$\Rightarrow$ Multiply both sides by x^{n-1}, $b + (\alpha_{n-1} + \alpha_{n-2} x + \ldots + \alpha_1 x^{n-2} + \alpha_0 x^{n-1}) = 0$

$\Rightarrow b = -(\alpha_{n-1} + \alpha_{n-2} x + \ldots + \alpha_1 x^{n-2} + \alpha_0 x^{n-1})$

$\because$ $\alpha_i \in A\ \forall i$ and $x \in A$, A: ring

$\therefore$ $b = x^{-1} \in A$

$\therefore$ A is a field.

"$\Leftarrow$": We know $\forall x \in A \setminus \{0\} \subseteq B$, $\exists\, x^{-1} \in A \subseteq B$ since A: field.

Consider $x \in B \setminus A$.

By def, $\exists\, f(T) \in A[T]$, f: monic + irr, s.t. $f(x) = 0$.

Say $f(T) = \sum_{i=0}^{n} \beta_i T^i$, $\beta_i \in A\ \forall i$, $\beta_n = 1$, $\beta_0 \neq 0$

Thus, $f(x) = 0 \Rightarrow x^n + \beta_{n-1} x^{n-1} + \ldots + \beta_1 x + \beta_0 = 0$

$\Rightarrow x(x^{n-1} + \beta_{n-1} x^{n-2} + \ldots + \beta_1) = -\beta_0 \in A$

As $-\beta_0 \in A$, thus $\exists\, c \in A$, s.t. $c = (-\beta_0)^{-1}$.

$\therefore$ $x[c(x^{n-1} + \beta_{n-1} x^{n-2} + \ldots + \beta_1)] = 1$

$\therefore$ $x^{-1} = c(x^{n-1} + \beta_{n-1} x^{n-2} + \ldots + \beta_1) \in B$

$\therefore$ By def, B is a field

$\therefore$ A: field $\Leftrightarrow$ B: field ✓

Notice, $A \subseteq B$ is an integral ext of rings

We know by def, $A/p = A/q \cap A \subseteq B/q$.

Moreover, $\forall b \in B$, we know $\exists\, f(T) \in A[T]$, f: monic + irred, s.t. $f(T) = \sum_{i=0}^{n} \alpha_i T^i = 0$, $\alpha_i \in A\ \forall i$, $\alpha_n = 1$, $\alpha_0 \neq 0$.

Then, for $\bar{b} \in B/q$, we have $\bar{f}(T) := \sum_{i=0}^{n} \bar{\alpha}_i T^i \in A/q \cap A[T] = A/p[T]$, s.t. $\bar{f}(\bar{b}) = 0$

$\therefore$ $A/p \subseteq B/q$ is an integral ext of rings.

$\because$ field $\Rightarrow$ domain

Now, q: prime + max ideal of B $\Leftrightarrow$ B/q: domain + B/q: field $\Leftrightarrow$ A/p: field $\Leftrightarrow$ p: max ideal of A

$\therefore$ In conclusion, q is a max ideal of B iff p is a max ideal of A. $\square$